

21) Ensemble learning -

Bagging (Bootstrap Aggregation)

1. What is Ensemble learning?

Ensemble learning means using more than one model to solve a problem.

Example (Real life):-

Imagine you want to buy a phone.
You ask 5 friends.

If 4 say "Buy Samsung", you follow the majority.
→ This is ensemble learning.

Each friend = one model.

Majority vote = final prediction.

Why do we use multiple models?
Because:-

- A single model can make mistakes.

- A single model can overfit (memorize training data).

- Multiple different models combined give more stable & accurate results.

Ensemble = "Teamwork of models"

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testing error varies significantly on different training sets.

Q. Why Only One Model is Not Enough?
⇒ A single model often suffers from:-

(A) High Variance

- Model learns noise
- Model changes too much with small changes in data.
- Common in Decision Trees

(B) Overfitting:-

- learns useless patterns
- performs well on training data, bad on test data.

Ensemble → reduces variance & overfitting

3. What is Bagging?

Bagging is one of the simplest & most famous ensemble methods.

Bagging = Bootstrap + Aggregation

4. Bootstrap (Sampling Step) → Have same model but on different forms (trained on different datasets)

Bootstrap = Random Sampling with Replacement

Meaning of "with replacement" ✓

✓ { You pick one sample randomly →
You put it back into the dataset →
So, it can be picked again.

Why do we do this?

• To create many different datasets from the original dataset.

Example:-

Original dataset = 100 samples

We want to create 3 bootstrapped datasets:

- Dataset 1 → 100 samples chosen randomly with replacement
- Dataset 2 → 100 samples chosen randomly with replacement
- Dataset 3 → 100 samples chosen randomly with replacement.

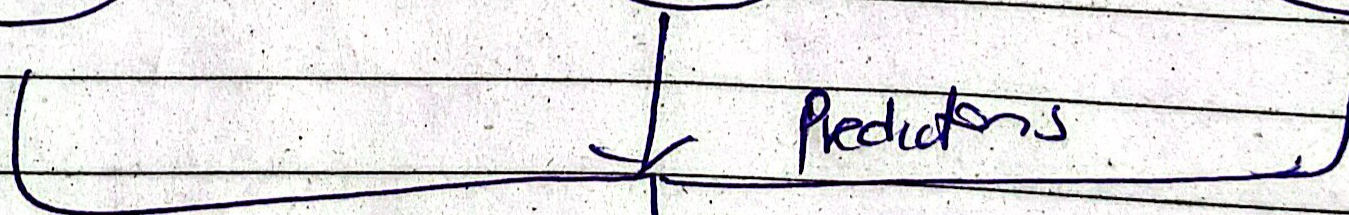
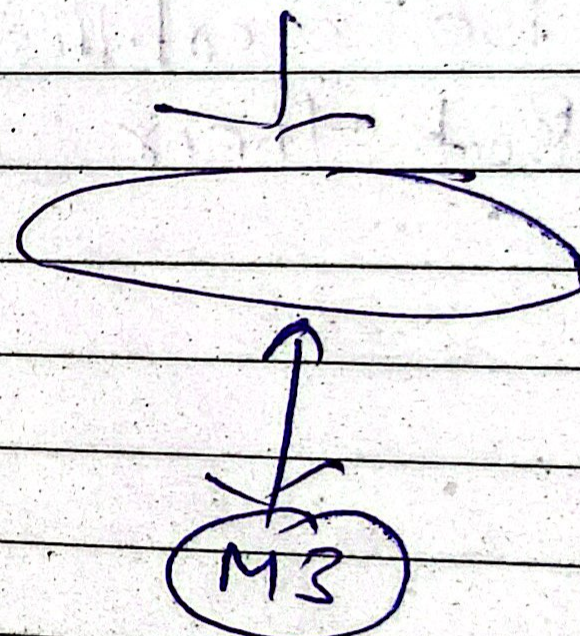
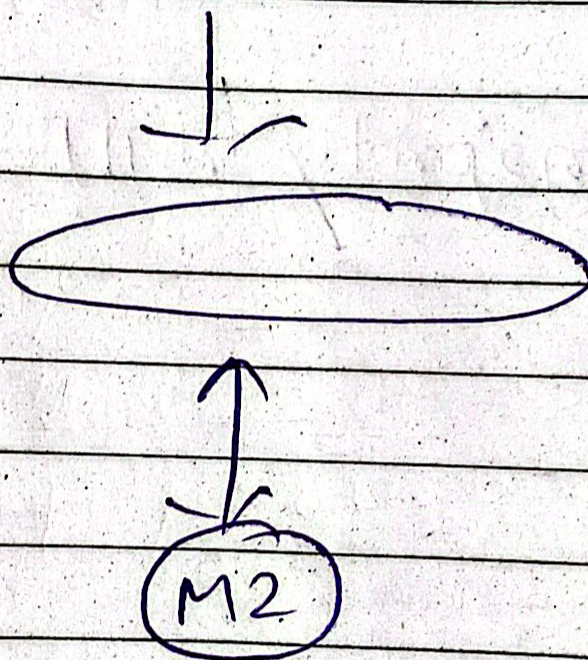
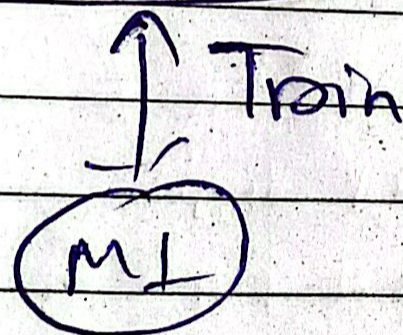
In each dataset:-

- Some samples will repeat
- Some samples may not appear at all
- That's normal & important.

100 Sample,

Bootstrap

70 sample



Aggregation

Average
Majority vote

M1, M2, & M3 are weak learners, because they are trained on smaller dataset. So, they won't be overfitted.

But their majority will be stronger.

→ Dataset can be nature of overfitting.
→ And bandwidth also.

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5. Train Many Weak Models (Learners)

After creating bootstrapped datasets:-

- Train Model 1 on Dataset 1 ✓
- Train Model 2 on Dataset 2 ✓
- Train Model 3 on Dataset 3 ✓
- ... 2nd so on.

All models use the same algorithm.

Examples:-

- All Decision Trees ✓
- All Logistic Regression models ✓
- All ANN models ✓

But because each model gets different data, they'll learn slightly different things.

Then diversity makes the ensemble strong.

6. Aggregation (Combining Results)

(Average)
After training many models, we combine their outputs.

for Classification:-

Uses majority voting ✓
→ class with highest votes = final answer ✓

for Regression:

Use Averaging ✓
→ Output = mean of all predictions ✓

7. Why Bagging works? ✓

Because:-

1. Reduces Variance ✓

Each model sees different data. ✓

Their errors cancel each other out. ✓

2. Reduces Overfitting

A single decision tree easily overfits
But 100 trees combined generalize better. ✓

3. More Stable

Small changes in data do not affect the final result much. ✓

Bagging converts weak learners → strong learner ✓

21 columns
also
random
columns

town	area	bedrooms	bathroom	plot	school rating	basement	price
monroe	2600	2	2	8500	8.1	Yes	55k
monroe	3000	3	3	9200	8.7	Yes	56.5k
monroe	3200	3	3	8750	8.7	No	61k
monroe	3600	4	4	10200	8.7	Yes	68k
monroe	4000	5	4	15000	8.1	Yes	72.5k
west candler	2600	2	2	7000	9.5	No	58.5k
west candler	2800	3	3	9000	9.5	No	61.5k
west candler	3300	3	4	10000	9.1	Yes	65000
west candler	3600	4	4	10500	9.1	Yes	71000
west candler	3100	3	3	7500	9.5	Yes	75k

town	area	school rating	basement	price
west candler	2800	9.5	No	61.5k
monroe	3600	8.7	Yes	68k
west candler	2600	9.5	No	58.5k
west candler	2800	9.5	No	61.5k
monroe	4000	8.1	Yes	72.5k
west candler	3100	9.5	Yes	75k

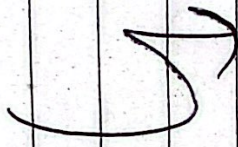
550k
Recum Tree
AVERAGE
52k

town	bathroom	plot	school rating	price
-	-	-	-	-

town	area	bedrooms	basement	price
-	-	-	-	-

Recum Tree
3
50k

Bagging



underlying model can be anything.
(svm, knn, logistic regression,
etc)

Bagged
Trees



Each model is a tree

Special type of

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8. Random forest (Special type of Bagging)

Random forest =

Bagging Decision Trees + Random feature selection

Steps:-

1. Bootstrap rows (like Bagging)
2. Randomly select only some features for each tree.
3. Train one decision tree
4. Repeat many times
5. Combine predictions of all trees

Why add random feature selection?

So trees become more different from each other
→ better performance.

Difference:

Method	What it uses
Bagging	Any algorithm (kNN, SVM, Tree, etc)
<u>Random forest</u>	Only Decision Trees + Column Sampling

Random forest is one of the most widely used ML algorithms.

9. Out-of-Bag Score (OOB Score)

When we do bootstrap sampling -

- Some samples are not selected
- These samples are called Out-of-Bag samples.

These samples act as mini test data.

Why useful?

- gives internal accuracy estimation ✓
- No need for separate test set ✓
- Works only with bagging-based models ✓

10. What Happens in Code (Concept Only)

Here is the step-by-step flow:-

Step 1:- Load Dataset
use pandas to read CSV file

Step 2:- Basic Checks

- `df.isnull().sum()` ✓
- `df.describe()`
- `df.outcome.value_counts()`

`df[outcome]`

Step 3:- Make features and labels

- $X \rightarrow$ input columns ✓
- $Y \rightarrow$ output / target (e.g. Outcome) ✓

Step 4:- Scaling

Use StandardScaler or MinMaxScaler to
normalize numeric columns ✓

Step 5:- Train-Test Split $\rightarrow$ test_size = 0.2 if dataset is balanced ✓

Step 6:- Train Models

Train:

- Single Decision Tree ✓
- Bagging Classifier ✓
- Random Forest Classifier ✓

Step 7:- Compare Scores

Use:-

- accuracy score ✓
- cross_val_score ✓

Usually:-

- Decision Tree has lowest accuracy
- Bagging gives higher
- Random forest gives highest

7.1. final Important points.

- Bagging reduces variance ✓
- Multiple weak models $\rightarrow$ strong model ✓
- Works best with unstable models
(like decision trees) ✓
- Random forest = Bagging + random feature sampling ✓
- OOB score = internal validation ✓
- Bagging works for both classification & regression. ✓